

Stress Prediction from Wearable Sensor Data

MSDS692 Project Review

Project Details:

This project aims to predict stress levels using physiological and motion data collected from wearable devices. Stress detection is a central focus in health informatics, as it enables proactive management of personal well-being, supports safer workplace environments, and provides value for clinical monitoring. The goal is to build machine learning and deep learning models that classify stress versus non-stress states and to evaluate which physiological parameters most strongly indicate stress responses (Schmidt et al., 2018).

The datasets used in this project are multimodal and complex, including electrocardiogram (ECG), electrodermal activity (EDA), respiration, temperature, and accelerometer signals. Each dataset varies in labeling schemes, sampling rates, and signal modalities, requiring careful preprocessing and feature extraction. This multimodal and heterogeneous nature of the data provides both opportunities and challenges: it allows richer modeling of stress responses but also demands tailored strategies for handling label imbalance, temporal dependencies, and noise in physiological signals (Shaffer & Ginsberg, 2017; Benedek & Kaernbach, 2010).

Project Status:

	Summary of the Work Planned for the Week	% Complete
Week 1	Submit project proposal, finalize topic, create initial plan.	100%
Week 2	Dataset selection (WESAD, HRV Sleep, PhysioNet), environment setup in Colab/Drive.	100%
Week 3	Data cleaning, preprocessing, feature extraction (EDA, HRV metrics, accelerometer features).	100%
Week 4	Implement baseline machine learning models (Logistic Regression, Random Forest) for supervised datasets; exploratory evaluation.	100%

Week 5	Develop and test deep learning models (LSTM, CNN, Autoencoder); handle class imbalance with resampling/oversampling.	70%
Week 6	Hyperparameter optimization, multimodal fusion across signals, refine deep learning models.	0%
Week 7	Final evaluation, visualizations, dashboard creation.	0%
Week 8	Final report writing, formatting, and presentation.	0%

Code Base

I plan to deliver my code base through a GitHub repository. Starting this week, I will begin creating and structuring the repository to ensure all scripts, notebooks, datasets (links), and documentation are clearly organized. The repository will include separate folders for raw and processed data, preprocessing scripts, model development (baseline and deep learning), and evaluation results. A detailed README.md file will also be provided to guide reviewers through the project workflow.

The finalized GitHub repository link will be submitted as a text document in the Final Deliverables folder in WorldClass by Sunday, October 19, 2025, at 11:55 p.m. MT.

Online Presence:

For my project's online presence, I will be using a GitHub repository as the central hub for my work. The repository will be structured with clear folders for preprocessing, modeling, evaluation, and results, along with a concise write-up that explains my overall process. In this write-up, I will highlight the reasoning behind my choices (who/what/why/when/how), the challenges I faced (such as class imbalance, windowing decisions, and balancing oversampling vs. under-sampling), and how I overcame them. I will also reflect on what I learned while building and testing models across different datasets, particularly the trade-offs between accuracy, F1-score, and generalization.

To make the repository more valuable, I will include well-documented code, inline explanations, and short commentary files that walk through my decision-making step by step. This will not only show my technical implementation but also my problem-solving strategies and critical thinking. I will also document failed approaches or roadblocks I encountered, since they are part of the learning journey. By the end of this week, I will finalize the structure, document the code, and prepare the repository for a 1-on-1 review. The final deliverable will be the GitHub repository link, submitted in the Final Deliverables folder in WorldClass by Sunday, October 19, 2025, at 11:55 p.m. MT.

Presentation:

All presentations should be between 10 and 15 minutes in length. The expectation is that you will produce a PowerPoint slide deck as the basis of your presentation. All presentations need to be finalized by Thursday, October 16, 2025. MSDS692_S41 - You will present live in class on October 16th. I will randomly assign the presentation order. Please ensure that you are in the classroom by 6:00 pm MT, logged into the Zoom session, and ready to present.

(I will prepare the presentation toward the end of Week 7 or early Week 8, after completing the project. It will be 10–15 minutes, designed to be engaging, conversational, and story-driven, focusing on why I chose the project, key insights, challenges, and outcomes. I will minimize technical details, instead highlighting decisions, learning, and practical takeaways. The presentation will acknowledge challenges, encourage audience interaction, and include Q&A. Each slide will tell a part of the project story, linking problem, methods, results, and conclusions, with proper references for all work. The goal is to convey the project journey and impact clearly and memorably.)

Acknowledgement:

- ☒ By checking this box, you acknowledge that you have read and understand all the information within this document. Late submissions will not be accepted.